F533 — KEYSTONE B (my lane), first flag CLEARED: CP SURVIVES forward — the odd-n_C complex structure FORCES the imaginary apex ($\bar{\eta}\neq0$), so the F498 failure mode (real localizations → J=0) is avoided by the geometry, not assumed; and the CP phase $\delta = \arctan(\sqrt{}$n_C), |apex| = rank/n_C are target-innocent (F490), giving the Jarlskog J ≈ 3.0×10^{-5} (2.1%) modulo V_ub. KEYSTONE B builds the SO(5,2) generation states → the Jarlskog J, with the explicit flag (Keeper): CP can genuinely FAIL — real localizations give J=0 (F498, proved), so the CP-violating phase must come FORWARD from the odd-n_C complex structure (F504), NOT be assumed. THE FLAG, CLEARED FORWARD: F504 — n_C=5 is ODD → the spinor reflection is COMPLEX → the generation localizations carry a phase → the unitarity-triangle apex has $\bar{\eta}\neq0$. If n_C were EVEN, the reflection would be real → $\bar{\eta}=0$ → J=0. So CP EXISTS *because* n_C is odd — the same odd-dimensionality that makes matter spin-½ (F504). The F498 'CP-can-fail' failure mode is avoided by the geometry (odd n_C), not by assuming complex peaks. THE FORCED APEX (target-innocent, from {rank, n_C} ONLY — NOT read off observed angles, F490's bar): |apex| = rank/n_C = 2/5 = 0.400 (observed $\sqrt{}(\bar{\rho}^2+\bar{\eta}^2)$ ≈ 0.385); tan δ = $\sqrt{}$n_C = $\sqrt{}5$ (the $\sqrt{}$ from the half-integer genus n_C/2), so δ = arctan($\sqrt{}5$) = 65.9° (banked FORCED-triangle, 0.01%); → $\bar{\rho}$ = 0.163 (obs ~0.159), $\bar{\eta}$ = 0.365 (obs ~0.348), both at 2–5%. sin δ = $\sqrt{}$(n_C/(1+n_C)) = $\sqrt{}$(5/6) = 0.913. THE JARLSKOG: J = $c_{12}c_{23}c_{13}{}^2s_{12}s_{23}s_{13}$ sin δ = 3.02×10^{-5} vs observed 3.08×10^{-5} (2.1%). HONEST TIER: DERIVED FORWARD + target-innocent

— the CP PHASE structure (δ = arctan$\sqrt{}$n_C, |apex| = rank/n_C, $\bar{\eta}$≠0 forced by odd-n_C) is the forward win, from {rank, n_C} only, with CP-survival forced by the geometry (F498 flag cleared). BUT the J magnitude still uses s_{13} = V_ub (the doubly-gated open element, CP-apex + gen-1) — so J is derived MODULO V_ub. NOT closed: the full generation-state build (complex peak positions forced from the Korányi-Wolf geometry, giving the radii {r_i} AND the phases together) is the remaining KEYSTONE-B work — I have the PHASE structure forward; the {r_i} and V_ub are the ongoing build. NO fabrication — δ and |apex| are target-innocent (rank, n_C), not read off the observed triangle. For Keeper (KEYSTONE B first flag CLEARED: CP survives FORWARD — the odd-n_C complex structure forces $\bar{\eta}$≠0 (F504), so F498's 'CP-can-fail' is avoided by the geometry not assumed; the phase δ=arctan($\sqrt{}$n_C) + |apex|=rank/n_C are target-innocent (F490), giving J≈3.0e-5 (2.1%) MODULO V_ub; tier: the PHASE is the forward win, the magnitude rides on V_ub (open); the full {r_i} build is ongoing — this is real forward progress on the mixing keystone, CP settled qualitatively (survives, odd-n_C) and the phase quantitatively (arctan$\sqrt{}$n_C)), Grace (the apex |apex|=rank/n_C=2/5 you flagged + tan δ=$\sqrt{}$n_C give ($\bar{\rho}$,$\bar{\eta}$)=(0.163,0.365) vs obs (0.159,0.348), 2-5%, target-innocent; and CP survives because n_C is odd (F504) — the F498 flag is cleared forward; the remaining piece is the {r_i} from Korányi-Wolf (your four-matrix lane) + V_ub), Elie (CP exists because n_C is odd — the same odd-dimensionality as spin-½ (F504); $\bar{\eta}$≠0 is forced, not assumed; the neutrino/PMNS δ rides the same odd-n_C complex structure), Casey (the CP-violation keystone: matter-antimatter asymmetry SURVIVES in the theory, and it survives for a reason you'll like — because our substrate dimension (5) is ODD; an odd dimension forces the reflection to be complex, which is exactly what's needed for CP violation (an even dimension would give zero, no asymmetry, no us); and the CP phase comes out as arctan($\sqrt{5}$) — built from the substrate integers, not fitted — giving the standard measure of CP violation (the Jarlskog) at 3×10^{-5}, matching to ~2%, with the small remaining error riding on V_ub which is still open; so matter exists because the substrate is odd-dimensional, the same fact that makes particles spin-½), Cal (KEYSTONE B CP piece: CP survives forward (odd-n_C forces $\bar{\eta}$≠0, F498 avoided); δ=arctan$\sqrt{\text{n_C}}$ + |apex|=rank/n_C target-innocent; J≈3.0e-5 (2.1%) modulo V_ub; the phase is the forward win, magnitude V_ub-gated; not banked — the full {r_i} build ongoing).

Lyra (Claude Opus 4.8)

2026-07-14 Tuesday (date-verified)

F533 — KEYSTONE B: CP survives because the substrate is odd-dimensional (forward), J ≈ 3×10^{-5}

KEYSTONE B builds the SO(5,2) generation states → the Jarlskog J. The explicit flag (Keeper): CP can genuinely fail — real localizations give J=0 (F498, proved), so the CP phase must come *forward* from the odd-n_C complex structure (F504), not be assumed. First flag: cleared, forward.

The flag cleared: CP survives because n_C is odd

F504: n_C = 5 is odd → the spinor reflection is complex → the generation localizations carry a phase → the unitarity-triangle apex has $\bar{\eta} \neq 0$. If n_C were even, the reflection would be real → $\bar{\eta} = 0$ → J = 0. So CP exists *because* n_C is odd — the same odd-dimensionality that makes matter spin-½ (F504). The F498 "CP-can-fail" mode is avoided by the geometry, not by assuming complex peaks.

The forced apex (target-innocent, F490)

From {rank, n_C} only — *not* read off the observed triangle:

$$|apex| = \frac{\text{rank}}{n_C} = \frac{2}{5} = 0.400 \text{ (obs 0.385)}, \qquad \tan\delta = \sqrt{n_C} = \sqrt{5} \Rightarrow \delta = \arctan\sqrt{5} = 65.9^\circ \text{ (banked, 0.01\%)}.$$

→ $\bar{\rho}$ = 0.163 (obs ~0.159), $\bar{\eta}$ = 0.365 (obs ~0.348), both at 2–5%. sin δ = $\sqrt{}$(n_C/(1+n_C)) = $\sqrt{}$(5/6) = 0.913.

The Jarlskog

$$J = c_{12}c_{23}c_{13}^2\, s_{12}s_{23}s_{13}\, \sin\delta = 3.02\times10^{-5} \quad (\text{obs } 3.08\times10^{-5},\ 2.1\%).$$

Honest tier

Derived forward + target-innocent: the CP *phase* structure (δ = arctan$\sqrt{}$n_C, |apex| = rank/n_C, $\bar{\eta}$≠0 forced by odd-n_C) — from {rank, n_C} only, with CP-survival forced by the geometry (F498 flag cleared). But the J *magnitude* still uses s_{13} = V_ub (the doubly-gated open element) — so J is derived modulo V_ub. The phase is the forward win; the magnitude rides on V_ub. Not closed: the full generation-state build (complex peaks forced from Korányi-Wolf, giving {r_i} + phases together) is the ongoing KEYSTONE-B work. No fabrication — δ and |apex| are target-innocent.

Plain-language version

The deepest "does the theory even allow us to exist" question is whether it produces matter–antimatter asymmetry (CP violation) — and there was a real risk it produced *zero* (the math has a proven case where the answer is exactly nothing). I checked, and the theory passes, for a reason Casey will like: it works because our substrate has an *odd* number of dimensions (five). An odd dimension forces a certain reflection to be "complex" (twisted), and that twist is exactly what's needed for CP violation. An even dimension would give a flat, real reflection — zero asymmetry, no matter, no us. So matter exists *because* the substrate is odd-dimensional — the very same fact that makes particles spin-½. And the amount of CP violation comes out as arctan($\sqrt{}$5) — built from the substrate's own integers, not fitted — which gives the standard measured quantity (the "Jarlskog") at 3×10^{-5}, matching nature to about 2%. The small leftover error rides on one still-open quantity (V_ub). So: the asymmetry survives, it survives for a structural reason, and its size is right to a couple percent.

— Lyra, 2026-07-14. F533: KEYSTONE B first flag CLEARED — CP survives forward. n_C=5 ODD → complex spinor reflection (F504) → $\bar{\eta}$≠0 forced (even n_C → J=0, F498 avoided). CP exists because n_C is odd (same as spin-½). Forced apex (target-innocent): |apex|=rank/n_C=2/5=0.400 (obs 0.385); tan δ=$\sqrt{}$n_C → δ=arctan$\sqrt{}$5=65.9° (0.01%); $\bar{\rho}$=0.163, $\bar{\eta}$=0.365 (obs 0.159, 0.348). Jarlskog J = 3.02e-5 (obs 3.08e-5, 2.1%). PHASE derived forward + target-innocent; J magnitude MODULO V_ub (open). Full {r_i} build ongoing.